

# An Unofficial Compendium of Worked Solutions to the Problems in *The Method of Coordinates*

by I.M. Gelfand, E. G. Glagoleva & A. A. Kirillov

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## Abstract

This book is an independent and unofficial collection of worked solutions to problems inspired by the book *The Method of Coordinates* by I. M. Gelfand, E. G. Glagoleva and A. A. Kirillov, first published in 1967. The original text presents a wide range of problems in elementary algebra, some accompanied by solutions and others left as exercises for the reader.

The aim of this book is to provide clear, logically consistent solutions to all problems addressed, written in a style accessible to motivated students and self learners. Proofs have been verified primarily by hand, and in some cases supported by small computational checks written in Scheme (a dialect of Lisp) or in Python. Large language models were used only as auxiliary tools, primarily for generating figures in L<sup>A</sup>T<sub>E</sub>X and for reviewing exposition.

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## Copyright and Disclaimer

### Unofficial companion.

This document is an **independent, unofficial collection of worked solutions** to problems from *The Method of Coordinates* by I. M. Gelfand , E. G. Glagoleva and A. A. Kirillov (first published 1967).

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The original problem statements are not reproduced here. All exposition and solutions are written independently by the author of this document.

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Project repository:

<https://github.com/deepak-venkatesh/gelfand-method-coordinates>

*For*  
*My Daughter and Son*

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## Working Notes

*Challenging Problems for Middle Schoolers:*

Problem Numbers:

*Typographical errors in:*

Problem Numbers:

## Part I

# Part 1

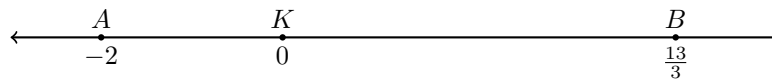
## 1 Chapter 1: The Coordinates of Points on a Line

### 1.1 The Number Axis

#### Exercise 1

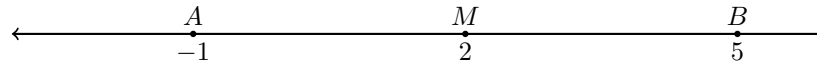
(a)

The points are marked below. Note that  $\frac{13}{3}$  lies somewhere between 4 and 5.



(b)

The two points are 3 units to the right and 3 units to the left of  $M$  as shown in the line below.





### 1.2 The Absolute Value of a Number

### 1.3 The Distance Between Two Points

## 2 Chapter 2: The Coordinates of Points in the Plane

### 2.1 The Coordinate Plane

### 2.2 Relations Connecting Coordinates

### 2.3 The Distance Between Two Points

### 2.4 Defining Figures

### 2.5 We Begin to Solve Problems

### 2.6 Other System of Coordinates

## 3 Chapter 3: The Coordinates of a Point in Space

### 3.1 Coordinate Axes and Planes

### 3.2 Defining Figures in Space

## Part II

# Part 2

## 4 Chapter 1: Introduction

### 4.1 Some General Considerations

### 4.2 Geometry as an Aid in Calculation

### 4.3 The Need for Introducing Four-Dimensional Space

### 4.4 The Peculiarities of Four-Dimensional Space

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### 4.5 Some Physics

GitHub repository:  
<https://github.com/deepak-venkatesh/gelfand-method-coordinates>

## 5 Chapter 2: Four-Dimensional Space

### 5.1 Coordinate Axes and Planes

### 5.2 Some Problems